

MongoDB — A case study



MongoDB is a NoSQL, Document-based model, and Schemaless Database. It is one of the most widely used database models. It is flexible, scalable, easy to maintain, and manage.

Before going deep dive into MongoDB. first, let us understand what is data model, the difference between the NoSQL and SQL, different types of NoSQL databases.



Now that we know MongoDB is a NoSQL database let us go into what is NoSQL and its need for it?

NoSQL stands for Not Only SQL. This means it is not based on SQL,

It has a dynamic schema for unstructured data. Data is stored in many ways which means it can be document-oriented, column-oriented, graph-based, or organized as a KeyValue store. This flexibility means that documents can be created without having a defined structure first.

Types Of NoSQL Databases

- Document Databases [[MongoDB](#), [CouchDB](#)]
- Column Databases [[Apache Cassandra](#)]
- Key-Value Stores [[Redis](#), [Couchbase Server](#)]
- Cache Systems [[Redis](#), [Memcache](#)]
- Graph Databases [[Neo4J](#)]

What is Data Model?

A data model is an abstract model that organizes elements of data and standardizes how they relate to one another and to the properties of real-world entities. It refers to the logical inter-relationships and data flow between different data elements involved in the information world.

There are many kinds of data models of which these are some of the models,

- Hierarchical database model
- Relational model
- Network model
- Object-oriented database model
- Entity-relationship model
- Document model

Every data can be categorized into 2 types SQL based and NoSQL-based.

SQL stands for Structured Query Language. It is designed for managing data in a relational database management system (RDBMS). It has a fixed schema, relational-based, Horizontally scalable, table-based model.

The main difference between SQL and NoSQL is as follows,

SQL	NoSQL
RELATIONAL DATABASE MANAGEMENT SYSTEM (RDBMS)	Non-relational or distributed database system.
These databases have fixed or static or predefined schema	They have have dynamic schema
These databases are not suited for hierarchical data storage.	These databases are best suited for hierarchical data storage.
These databases are best suited for complex queries	These databases are not so good for complex queries
Vertically Scalable	Horizontally scalable
Follows ACID property	Follows CAP(consistency, availability, partition tolerance)

MongoDB: —

MongoDB uses the Documental model as its core. It uses JSON to store the data. In the SQL world, we have DB → Table → Fields and records, whereas in MongoDB we have DB → Collections → Documents.

Each data is stored in documents, due to which it has the capacity to add as many columns to the data.

Its main features include

1. Ad-hoc queries
2. Indexing
3. Mapreduce
4. Aggregation
5. Replication
6. Load balancing

These make using MongoDB a great choice for MySQL. Coming to how the industry is using MongoDB there are a number of ones who use namely Google, KPMG, Intuit, Adobe, Forbes, Toyota, AstraZanika, Vivint, Royal Bank of Scotland, SAP, EA, Cisco, Appolo Group, Pearson, and many more.

What makes MongoDB a different and better database ?

Well , MongoDB is quite different when compared to the normal databases in prevailing. Generally SQL and NoSQL databases. MongoDB falls under the category of **NoSQL databases** as it does not maintain any feature information (or) metadata , technically called as “**schema**”. Also , it stores the data row wise other than a typical way of column wise storage with fields specified. Each row of data (record) is called a document which makes it the primary element of the huge database. Multiple documents make up a collection which when combined make up the database. Thus MongoDB is also called as a **document oriented database**.

Case study — 1 AstraZeneca



AstraZeneca embarked on an ambitious program to use next-generation genome sequencing to develop drugs to fight all kinds of diseases, including cancer. The technology creates a synthetic version of messenger RNA, which helps create protein in cells. If successful, the proteins could fight cancer, among other diseases.

Unfortunately, such genomic sequencing requires a great deal of computing power. As Jason Tetrault, architect of R&D information at AstraZeneca explained recently, analyzing 88 whole human genomes took 15,000 hours and 171 terabytes (TB) of data. Analyzing a single human genome can take four days.

AstraZeneca’s experiment involved taking 10% of all its compounds and pulling in information from its disparate database systems. Using MongoDB, the company was able to execute Tanimoto comparisons on about 500,000,000 compounds.

AstraZeneca can pursue links and patterns that it never noticed before. “Maybe this worked for 10% in liver cancer but wow this lung cancer actually has the same biomarker says, Tetrault.

Case study — 2 Forbes

Forbes

In just six months, Forbes migrated its platform to Google Cloud and MongoDB Atlas. Results include:

- 58% faster build time for new products and fixes
- Accelerated release cycle by 4x
- Reduced total cost of ownership by 25%
- 28% increase in subscriptions from new newsletters

During the pandemic the cloud infrastructure has also helped the website scale to an extraordinary number of users and helped the team stay nimble, introducing and testing a number of new features.

From June to December last year 2020, traffic continued to grow — setting new records month after month. Then came COVID-19. Like many high-profile publications, Forbes’ coverage of the pandemic has driven a further increase in traffic, reaching record traffic in May with more than 120 million unique visitors.

The combination of MongoDB Atlas within Google Cloud’s native microservice architecture would also prove a wise choice. Firstly, Google Cloud’s Kubernetes Engine made it more manageable to orchestrate Forbes’ horde of more than 50 microservices. (Focused, self-contained codebases that allow each service to be easily understood, modified quickly without dependencies on other services, and to be built with the best technology for the task.) Secondly, Atlas was also able to work seamlessly with Google Cloud’s suite of services to build even more powerful tools for its writers.

In April, Forbes introduced a trending story recommendation engine for journalists. The engine scrapes the internet for trending stories and uses Google Cloud’s machine learning to make suggestions to appropriate contributors, either through a Slack bot or through the custom CMS (the CMS itself had been rewritten again in 2019 to become more of an AI and

analytics-driven platform). Like everything at Forbes, the engine is still undergoing heavy testing but it has proved instantly helpful to Forbes' editorial team and contributor network.

Forbes' cloud migration led to a 58% improvement in build speed and the release cycle has improved 2x to 10x (depending on the service). Other efficiencies that came with the migration resulted in a 25% reduction in the total cost of ownership.

→ **Toyota :**

Toyota is famous for the quality of its vehicles. Now that same standard is being applied to how its development teams create Internet of Things software that will enable the move to smart, autonomous and safer factories of the future. Underpinning this change is *MongoDB Atlas*, MongoDB's fully-managed, global cloud database service, which Toyota Material Handling runs on Microsoft Azure.



"The most beautiful part is the data model. Everything is a natural JSON document. So for the developers, it is easy, really easy for them to work with quickly. Spending time on building business value, rather than data modeling."

But it wasn't just the data model. Toyota Material Handling had eight evaluation criteria that were used to decide on which database platform they would use to make the switch which were ;

1. Performance
2. Automatic scalability , operations and maintenance
3. Security and compliance to safeguard sensitive business
4. Data locality

5. Automatic backups and restore
6. Cloud Agnostic
7. Developer friendly
8. High productivity

→ **Intuit :**

[Intuit](#) empowers small businesses, accountants, and individuals with tax preparation and financial software. In addition, the company enables small businesses to build websites without any technical expertise through its Intuit Websites arm. In order to recommend conversion and lead generation improvements to their customers, Intuit needed a way to collect and analyze data from these sites. Initially, the company tried off the shelf products like Google Analytics and Omniture, but they realized that with more than 500,000 hosted websites and 10 years worth of user data, they required a simple, high-performance solution. With querying and map reduce functionality, MongoDB was the best platform to suit Intuit's needs.



Intuit found MongoDB to be 2.5 times faster for writes than MySQL and overall, superior to both relational and non-relational options.

Due to the distributed nature of their systems and sheer abundance of data, Intuit encountered difficulties when attempting to build their analytics service using a relational database like MySQL or Oracle. MongoDB provided a solution that was not only easy to deploy, but comparatively more efficient than the other relational and non-relational choices evaluated. Intuit spent a mere week developing a prototype analytics solution based on MongoDB.

Aadhar

Aadhar is an excellent example of real world use cases of MongoDB. In recent times, there has been some controversy revolving around CIA's non-profit Venture Capital arm, In-Q-Tel,

backing the company, which developed MongoDB. Putting aside the controversy, let's look at the MongoDB's role in Aadhar.

India's Unique Identification project, aka Aadhar, is the world's biggest biometrics database. Aadhar is in the process of capturing demographic and biometric data of over 1.2 billion residents. Aadhar has used MongoDB as one of its databases to store this huge amount of data. MongoDB was among several database products, apart from MySQL, Hadoop and HBase, originally procured for running the database search. Here, MySQL is used for storing demographic data and MongoDB is used to store images. According to techcrunch.com, MongoDB has nothing to do with the "sensitive" data.

Shutterfly

Shutterfly is a popular internet-based photo sharing and personal publishing company that manages a store of more than 6 billion images with a transaction rate of up to 10,000 operations per second. Shutterfly is one of the companies that transitioned from Oracle to MongoDB.

During the evaluation at the time of transition to MongoDB, it became apparent that a non-relational database would suit the Shutterfly's data needs better and thereby possibly improving programmer's productivity as well as performance and scalability.

Shutterfly considered a wide variety of alternate database systems, including Cassandra, CouchDB and BerkeleyDB, before settling on MongoDB. Shutterfly has installed MongoDB for metadata associated with uploaded photos, while for those parts of the application that require richer transactional model, like billing and account management, the traditional RDBMS is still in place.

Till now, Shutterfly is happy with its decision of transitioning to MongoDB and what Kenny Gorman (Data Architect of Shutterfly) has to say about it is, "I am a firm believer in choosing the correct tool for the job, and MongoDB was a nice fit, but not without compromises."

MetLife

MetLife is a leading global provider of insurance, annuities and employee benefit programs. They serve about 90 million customers and hold leading market positions in the United States, Japan, Latin America, Asia, Europe and the Middle East. MetLife uses MongoDB for "The Wall", an innovative customer service application that provides a consolidated view of MetLife customers, including policy details and transactions. The Wall is designed to look and function like Facebook and has improved customer satisfaction and call centre productivity. The Wall brings together data from more than 70 legacy systems and merges it into a single record. It runs across six servers in two data centres and presently stores about 24 terabytes of data. MongoDB-based applications are part of a series of Big Data projects that MetLife is working on to transform the company and bring technology, business and customers together.

eBay

eBay is an American multinational internet consumer-to-consumer corporation, headquartered in San Jose. eBay has a number of projects running on MongoDB for search suggestions, metadata storage, cloud management and merchandizing categorization.

MongoDB CRUD Operations

CRUD operations *create*, *read*, *update*, and *delete* [documents](#).

You can connect with driver methods and perform CRUD operations for deployments hosted in the following environments:

You can [perform CRUD operations in the UI](#) for deployments hosted in [MongoDB Atlas](#).

Create Operations

Create or insert operations add new [documents](#) to a [collection](#). If the collection does not currently exist, insert operations will create the collection.

MongoDB provides the following methods to insert documents into a collection:

- [db.collection.insertOne\(\)](#)
- [db.collection.insertMany\(\)](#)

In MongoDB, insert operations target a single [collection](#). All write operations in MongoDB are [atomic](#) on the level of a single [document](#).

```
db.users.insertOne(  ← collection
  {
    name: "sue",      ← field: value
    age: 26,          ← field: value
    status: "pending" ← field: value
  }                  } document
)
```

For examples, see [Insert Documents](#).

Read Operations

Read operations retrieve [documents](#) from a [collection](#); i.e. query a collection for documents. MongoDB provides the following methods to read documents from a collection:

- [db.collection.find\(\)](#)

```
db.users.find(  
  { age: { $gt: 18 } },  
  { name: 1, address: 1 }  
) .limit(5)
```



- ← collection
- ← query criteria
- ← projection
- ← cursor modifier

You can specify [query filters or criteria](#) that identify the documents to return.

For examples, see:

- [Query Documents](#)
- [Query on Embedded/Nested Documents](#)
- [Query an Array](#)
- [Query an Array of Embedded Documents](#)

Update Operations

Update operations modify existing [documents](#) in a [collection](#). MongoDB provides the following methods to update documents of a collection:

- [db.collection.updateOne\(\)](#)
- [db.collection.updateMany\(\)](#)
- [db.collection.replaceOne\(\)](#)

In MongoDB, update operations target a single collection. All write operations in MongoDB are [atomic](#) on the level of a single document.

You can specify criteria, or filters, that identify the documents to update. These [filters](#) use the same syntax as read operations.

```
db.users.updateMany(  
  { age: { $lt: 18 } },  
  { $set: { status: "reject" } }  
)
```



- ← collection
- ← update filter
- ← update action

Delete Operations

Delete operations remove documents from a collection. MongoDB provides the following methods to delete documents of a collection:

- [db.collection.deleteOne\(\)](#)
- [db.collection.deleteMany\(\)](#)

In MongoDB, delete operations target a single collection. All write operations in MongoDB are atomic on the level of a single document.

You can specify criteria, or filters, that identify the documents to remove. These filters use the same syntax as read operations.

```
db.users.deleteMany(  
  { status: "reject" }  
)
```



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